Sahaj Bhadja

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Summary

Quantitative developer pursuing an MSCS at USC with 2+ years productionizing **Python** data pipelines, **SQL/PostgreSQL** systems, and model execution workflows. Strong systems mindset for reliable ingestion, validation, monitoring, and repeatable outputs with interest in systematic investing and quantitative methods.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Junior Quantitative Developer *September 2025 – Present*
- Owned end-to-end **ETL-style pipelines** using **Python**, FastAPI, **PostgreSQL**, and Azure SQL, processing **5K+ monthly jobs** with incremental loads, reconciliation checks, and validation frameworks achieving **99%+ production reliability**.
- Built monitoring-ready data transformation and model execution workflows with logging, error handling, and CI/CD releases, optimizing SQL queries and indexing to reduce dashboard latency from **1.2s to 350ms** for auditability and repeatability.

Grad8 **Los Angeles, CA**
Quantitative Developer Intern *June 2025 – August 2025*
- Productionized **Python** grading pipelines processing **10K+ daily requests** with clean modular Flask services, unit-tested batch execution, and token-cost tracking libraries supporting repeatable research-to-production workflows for instructor-facing analytics.
- Implemented concurrent data-processing engines with structured logging and GitHub Actions CI/CD, troubleshooting production failures and delivering maintainable backend code aligned with reliability standards under peak load conditions.

USC AD Labs **Los Angeles, CA**
Quantitative Research Developer *January 2025 – May 2025*
- Deployed research models into scalable **Python** pipelines processing **100K+ time-series trajectory logs**, building simulation evaluation infrastructure and feature-generation workflows that aligned research prototypes with reproducible production-quality experimentation.
- Collaborated with researchers to productionize signal-aggregation and closed-loop **simulation** frameworks using PyTorch, improving long-horizon planning metrics by **18%** while maintaining disciplined version control and experiment logging practices.

Aasma Technology Solutions **India**
Quantitative Developer - (SecureSync) *August 2023 – July 2024*
- Built scalable **data ingestion** and transformation pipelines on **AWS Glue** and S3, automating JSON/CSV/Parquet normalization and improving ETL efficiency by **68%** with PyTest-validated releases and Jenkins monitoring across production environments.
- Troubleshot pipeline and API failures across distributed systems, implementing pagination, caching, and indexing optimizations that cut response times by **30%** while partnering with stakeholders to improve workflow automation and code maintainability.

LST Capitals **India**
Quantitative Developer Intern *November 2022 – January 2023*
- Engineered **Python** analytics pipelines on Databricks and Spark ingesting **50K+ booking and payment records**, aggregating revenue summaries and usage metrics for internal dashboards with sub-**200ms** API performance on financial transaction workflows.
- Developed Flask services integrating payment webhooks and coupon validation logic, writing testable production code with structured error handling that reduced manual refund processing by **70%** across multi-currency financial data systems.

Skills

Quant & Python: Python, NumPy, pandas, SciPy (familiar), CVXPY (familiar), statistics, linear algebra, optimization, time-series analysis, machine learning, backtesting, Monte Carlo simulation (familiar), portfolio analytics (familiar), financial markets
Data & Production: SQL, PostgreSQL, SQL Server, ETL, data pipelines, data validation, reconciliation, Airflow (familiar), Prefect (familiar), monitoring, logging, alerting, Git, PyTest, CI/CD, Jenkins, GitHub Actions, Docker, AWS, Azure
Engineering: FastAPI, Flask, REST APIs, production engineering, reliability, maintainability, testable code, problem solving, Agile, market data APIs (familiar), Bloomberg (familiar)

Projects

PortfolioLab Code | *Python, pandas, NumPy, CVXPY, Backtesting*
- Built a systematic portfolio research toolkit using **pandas** and **NumPy** for signal aggregation, constraint-based **optimization**, and **backtesting** workflows, with **Monte Carlo simulation** modules to evaluate outcome distributions and portfolio robustness.

FinNexus Live | *Financial Data, Plaid APIs, Transaction Analytics*
- Developed a fintech platform processing **1M+ transactions** with Plaid bank-linking APIs, real-time filtering, and secure tokenized access patterns, demonstrating scalable ingestion and analytics over high-volume **financial data** pipelines.

Linguistix | *Python, AWS, Docker, Production Pipelines, CI/CD*
- Deployed a containerized **Python** data-processing API on AWS ECS with CodePipeline CI/CD, validation checks, and automated releases, reflecting production engineering practices for reliable model-serving and pipeline operation at scale.